

Area, Perimeter and Volume

GCSE Maths · Worked Solutions

20 questions · 62 marks total

jtmaths.org · free maths resources

Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Work out the area of a rectangle measuring 7 cm by 5 cm.

1 Area of a rectangle = length $\times$ width

2 $7 \times 5 = 35$

Answer: 35 cm²

2 [2 marks]

Work out the area of a triangle with base 12 cm and perpendicular height 5 cm.

1 Area = $\frac{1}{2} \times \text{base} \times \text{height}$

2 $= \frac{1}{2} \times 12 \times 5 = 30$

Answer: 30 cm²

3 [2 marks]

Work out the area of a trapezium with parallel sides 6 cm and 10 cm and perpendicular height 4 cm.

1 Area = $\frac{1}{2} \times (a + b) \times h$

2 $a + b = 6 + 10 = 16$

3 $= \frac{1}{2} \times 16 \times 4 = 32$

Answer: 32 cm²

4 [2 marks]

Work out the circumference of a circle of radius 7 cm, correct to 1 decimal place.

1 Circumference = $2\pi r = 2 \times \pi \times 7 = 14\pi$

2 $14 \times 3.14159... = 43.982...$

Answer: 44.0 cm (1 d.p.)

5

[3 marks]

Work out the area of a circle of radius 6 cm, correct to 1 decimal place.

$$1 \text{ Area} = \pi r^2$$

$$2 = \pi \times 6^2 = 36\pi$$

$$3 36 \times 3.14159... = 113.097...$$

Answer: 113.1 cm² (1 d.p.)

6

[3 marks]

A cuboid measures 5 cm by 4 cm by 9 cm.

(a) Work out its volume.

(b) Work out its total surface area.

$$1 \text{ (a) Volume} = 5 \times 4 \times 9 = 180 \text{ cm}^3$$

2 (b) The six faces come in three matching pairs.

$$3 5 \times 4 = 20, \quad 5 \times 9 = 45, \quad 4 \times 9 = 36$$

$$4 \text{ Surface area} = 2 \times (20 + 45 + 36) = 2 \times 101 = 202 \text{ cm}^2$$

Answer: (a) 180 cm³ (b) 202 cm²

7

[3 marks]

Work out the volume of a cylinder with radius 3 cm and height 8 cm, correct to 1 decimal place.

$$1 \text{ Volume of a cylinder} = \pi r^2 h$$

$$2 = \pi \times 3^2 \times 8 = 72\pi$$

$$3 72 \times 3.14159... = 226.194...$$

Answer: 226.2 cm³ (1 d.p.)

8

[3 marks]

An L-shape is made by taking a rectangle 10 cm by 6 cm and cutting away a rectangle 4 cm by 3 cm from one corner.

Work out the area of the L-shape.

1 Work out the two areas and subtract.

$$2 \text{ Large rectangle: } 10 \times 6 = 60 \text{ cm}^2$$

$$3 \text{ Cut-out rectangle: } 4 \times 3 = 12 \text{ cm}^2$$

$$4 60 - 12 = 48 \text{ cm}^2$$

Answer: 48 cm²

9

[4 marks]

A semicircle has radius 5 cm.

(a) Work out its area, correct to 1 decimal place.

(b) Work out its perimeter, correct to 1 decimal place.

$$1 \text{ (a) Area of the whole circle} = \pi \times 5^2 = 25\pi = 78.539\dots \text{ cm}^2$$

$$2 \text{ Half of that is } 39.269\dots \text{ cm}^2$$

3 (b) The perimeter is the curved part **plus** the straight diameter.

$$4 \text{ Curved part} = \frac{1}{2} \times 2\pi \times 5 = 5\pi = 15.707\dots \text{ cm}$$

$$5 \text{ Diameter} = 10 \text{ cm, so the perimeter is } 15.707 + 10 = 25.707\dots \text{ cm}$$

Answer: (a) 39.3 cm² (b) 25.7 cm (both 1 d.p.)

10

[4 marks]

A prism has a trapezium cross-section with parallel sides 5 cm and 9 cm and perpendicular height 4 cm. The prism is 12 cm long.

(a) Work out the area of the cross-section.

(b) Work out the volume of the prism.

$$1 \text{ (a) Area} = \frac{1}{2} \times (5 + 9) \times 4$$

$$2 = \frac{1}{2} \times 14 \times 4 = 28 \text{ cm}^2$$

3 (b) Volume of a prism = area of cross-section $\times$ length

$$4 = 28 \times 12 = 336 \text{ cm}^3$$

Answer: (a) 28 cm² (b) 336 cm³

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

Work out the volume of a sphere of radius 3 cm. Give your answer in terms of π .

$$1 \text{ Volume of a sphere} = \frac{4}{3}\pi r^3$$

$$2 \text{ } r^3 = 27, \text{ so the volume is } \frac{4}{3} \times 27\pi = 36\pi$$

Answer: 36 π cm³

2

[2 marks]

Work out the volume of a cone with radius 4 cm and height 9 cm. Give your answer in terms of π .

$$1 \text{ Volume of a cone} = \frac{1}{3}\pi r^2 h$$

$$2 = \frac{1}{3} \times \pi \times 16 \times 9 = 48\pi$$

Answer: 48 π cm³

3

[3 marks]

Work out the surface area of a sphere of radius 5 cm, correct to 3 significant figures.

$$1 \text{ Surface area of a sphere} = 4\pi r^2$$

$$2 = 4 \times \pi \times 25 = 100\pi$$

$$3 100 \times 3.14159... = 314.159...$$

Answer: 314 cm² (3 s.f.)

4

[3 marks]

A sector of a circle has radius 8 cm and angle 45°.

Work out the arc length, correct to 3 significant figures.

$$1 \text{ The sector is } \frac{45}{360} = \frac{1}{8} \text{ of the whole circle.}$$

$$2 \text{ Full circumference} = 2 \times \pi \times 8 = 16\pi = 50.265... \text{ cm}$$

$$3 \text{ Arc} = \frac{1}{8} \times 50.265... = 6.2832... \text{ cm}$$

Answer: 6.28 cm (3 s.f.)

5

[4 marks]

A sector of a circle has radius 10 cm and angle 72°.

(a) Work out its area, correct to 1 decimal place.

(b) Work out its perimeter, correct to 1 decimal place.

$$1 \text{ The sector is } \frac{72}{360} = 0.2 \text{ of the whole circle.}$$

$$2 \text{ (a) Full area} = \pi \times 100 = 314.159... \text{ cm}^2$$

$$3 \text{ Sector area} = 0.2 \times 314.159... = 62.831... \text{ cm}^2$$

$$4 \text{ (b) Arc} = 0.2 \times 2\pi \times 10 = 0.2 \times 62.831... = 12.566... \text{ cm}$$

$$5 \text{ The perimeter also includes the two straight radii: } 12.566 + 10 + 10 = 32.566... \text{ cm}$$

Answer: (a) 62.8 cm² (b) 32.6 cm (both 1 d.p.)

6

[4 marks]

A cone has radius 5 cm and slant height 13 cm.

(a) Work out the curved surface area, correct to 3 significant figures.

(b) Work out the total surface area, correct to 3 significant figures.

$$1 \text{ (a) Curved surface area of a cone} = \pi rl$$

$$2 = \pi \times 5 \times 13 = 65\pi = 204.203... \text{ cm}^2$$

$$3 \text{ (b) The total also includes the circular base of area } \pi r^2 = 25\pi$$

$$4 \text{ Total} = 65\pi + 25\pi = 90\pi$$

$$5 90 \times 3.14159... = 282.743... \text{ cm}^2$$

Answer: (a) 204 cm² (b) 283 cm² (both 3 s.f.)

7

[4 marks]

A solid is made from a cylinder of radius 4 cm and height 10 cm, with a hemisphere of radius 4 cm on top.

Work out the total volume, correct to 3 significant figures.

$$1 \text{ Cylinder: } \pi r^2 h = \pi \times 16 \times 10 = 160\pi$$

$$2 \text{ Hemisphere: half of } \frac{4}{3}\pi r^3 = \frac{2}{3} \times \pi \times 64 = \frac{128}{3}\pi$$

$$3 \quad \frac{128}{3} = 42.666\dots, \text{ so the total is } (160 + 42.667)\pi = 202.667\pi$$

$$4 \quad 202.667 \times 3.14159\dots = 636.696\dots \text{ cm}^3$$

Answer: 637 cm³ (3 s.f.)

8

[4 marks]

A cylinder has volume 500 cm³ and height 10 cm.

Work out its radius, correct to 2 decimal places.

$$1 \text{ Volume} = \pi r^2 h, \text{ so } 500 = \pi \times r^2 \times 10$$

$$2 \quad r^2 = \frac{500}{10\pi} = \frac{50}{\pi}$$

$$3 \quad \frac{50}{3.14159\dots} = 15.9155\dots$$

$$4 \quad r = \sqrt{15.9155\dots} = 3.9894\dots \text{ cm}$$

Answer: 3.99 cm (2 d.p.)

9

[4 marks]

A sphere of radius r fits exactly inside a cylinder of radius r and height $2r$.

Show that the volume of the sphere is $\frac{2}{3}$ of the volume of the cylinder.

$$1 \text{ Sphere: } V = \frac{4}{3}\pi r^3$$

$$2 \text{ Cylinder: } V = \pi r^2 h \text{ with } h = 2r, \text{ so } V = \pi r^2 \times 2r = 2\pi r^3$$

$$3 \quad \text{Divide the sphere volume by the cylinder volume: } \frac{\frac{4}{3}\pi r^3}{2\pi r^3}$$

$$4 \quad \text{The } \pi r^3 \text{ cancels, leaving } \frac{4}{3} \div 2 = \frac{4}{6} = \frac{2}{3}$$

5 So the sphere occupies exactly two-thirds of the cylinder.

Answer: $\frac{4}{3}\pi r^3 \div 2\pi r^3 = \frac{2}{3}$

A solid is made from a cone of radius 6 cm and slant height 10 cm standing on top of a cylinder of radius 6 cm and height 15 cm.

(a) Work out the vertical height of the cone.

(b) Work out the total volume of the solid, to the nearest cm^3 .

(c) Work out the total surface area of the solid, to 3 significant figures. (The solid has a flat circular base but no join surfaces.)

1 (a) The radius, height and slant height form a right-angled triangle.

2 $\text{height}^2 = 10^2 - 6^2 = 100 - 36 = 64$, so the height is 8 cm

3 (b) Cylinder: $\pi \times 36 \times 15 = 540\pi$

4 Cone: $\frac{1}{3} \times \pi \times 36 \times 8 = 96\pi$, so the total is 636π

5 $636 \times 3.14159... = 1998.05...$, so 1998 cm^3

6 (c) Curved cone: $\pi rl = \pi \times 6 \times 10 = 60\pi$ · curved cylinder: $2\pi rh = 2\pi \times 6 \times 15 = 180\pi$ · base: $\pi r^2 = 36\pi$

7 Total = $60\pi + 180\pi + 36\pi = 276\pi = 867.08... \text{ cm}^2$

Answer: (a) 8 cm (b) 1998 cm^3 (c) 867 cm^2